

AI Research Intelligence Assistant

Ask questions across a company's research library and get answers grounded in the reports, with citations to the exact page.

✓ Working prototype built and running

The screenshot displays the AI Research Intelligence Assistant interface. On the left, a sidebar contains a 'New question' button, a 'LIBRARIES' section with 'Acme Corp', 'Globex', and 'Initech' (all checked), and a 'HISTORY' section with two recent queries. The main content area shows search results for 'BIS Annual Economic Report 2024'. It includes a list of bullet points about macroeconomic forecasting and financial market applications, a paragraph about the challenges of using AI in central banks, and a 'SOURCES' section. A search bar at the bottom of the main area contains the text 'Ask a question about the selected libraries...'. On the right, a preview of the 'BIS Annual Economic Report 2024' is shown, displaying text about the impact of AI on central banks and financial markets.

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1 Executive summary

The company's historical research holds a lot of value, but it is locked inside individual PDF reports. Customers have to open reports one by one to find information and connect it across years. This proposal puts forward an **AI research assistant**. An authorised user picks the research libraries they want, asks a question in plain language, and gets an answer built only from those reports. Every statement cites the report and page it came from.

The concept is **already built as a working prototype**, not just a design. Today it:

- ingests real PDF reports page by page, with an exact page index for every page
- finds relevant pages with hybrid search, combining meaning-based and keyword search
- writes answers with a three-agent pipeline: a Query Agent plans the search, an Answer Agent writes the cited answer, and an Evaluator Agent checks every claim against its source
- opens any citation in a built-in PDF viewer, at the cited page with the evidence highlighted
- restricts each user to the libraries they're authorised for, enforced in the database itself
- gives administrators control over documents, libraries and users, plus a quality dashboard

530

pages of BIS annual reports indexed in testing

100%

groundedness on the questions asked so far

19–28 s

end to end per question, including the evaluator check

69

automated tests passing (API, UI, database security)

Recommendation. Run a **10-week engagement with the client**, built around a **weekly meeting**, with 5–10 client users on 1–2 real research libraries. Measure time saved, citation accuracy and user satisfaction against the targets in section 13. Then decide whether to fund a production build. The prototype keeps the investment small until that value is proven.

2 The problem

The company has a substantial body of historical research, but it is published as PDF reports. That causes three problems:

- **Finding information is slow.** A user has to know which report to open, then search inside it. Questions that span several reports or years, such as "how has the view on X changed since 2022?", mean reading many documents.
- **Value goes unused.** The insight is there, but the format limits how easily customers can interrogate it. Older reports are rarely revisited.
- **An enterprise build is a big bet.** Committing to a full enterprise knowledge platform before the concept and its customer value are proven would need a much larger investment.

Desired outcome

A working prototype where an **authorised user can ask questions across a selected company's research library** and get useful, **source-grounded** answers they can verify. That lets the business demonstrate and assess the concept before deciding on a production build.

Design principles

PRINCIPLE	WHAT IT MEANS IN THE DESIGN
Grounded, not generative	The model may only use text retrieved from the selected reports. If the reports don't cover a question, the assistant says so rather than guessing.
Verifiable in one click	Every factual sentence carries a citation. Clicking it opens the PDF at the exact physical page, with the supporting sentence highlighted.
Secure by default	Access is checked in the database (row-level security), not only in the user interface. A user cannot retrieve text from a library they haven't been granted.
Small to start, easy to grow	Managed services (Supabase, Vercel, DeepSeek) mean no servers to run. Each component can later be swapped for an enterprise equivalent.

3 The solution: what users do

There are two roles. **Administrators** build the libraries; **users** ask questions and verify answers.

Administrator workflow

1. Create a **library** per company or research collection.
2. Upload PDF reports and tag them with one or more libraries. Indexing runs automatically and shows progress page by page.
3. Invite users and choose which libraries each one can search.
4. Monitor answer quality, feedback and usage on the Insights dashboard.

User workflow

1. Sign in, by invitation only.
2. Select the libraries to search: one company or several.
3. Ask a question in plain language. The answer streams in with citation chips such as `S1`.
4. Click any citation to open the source page, and rate the answer.

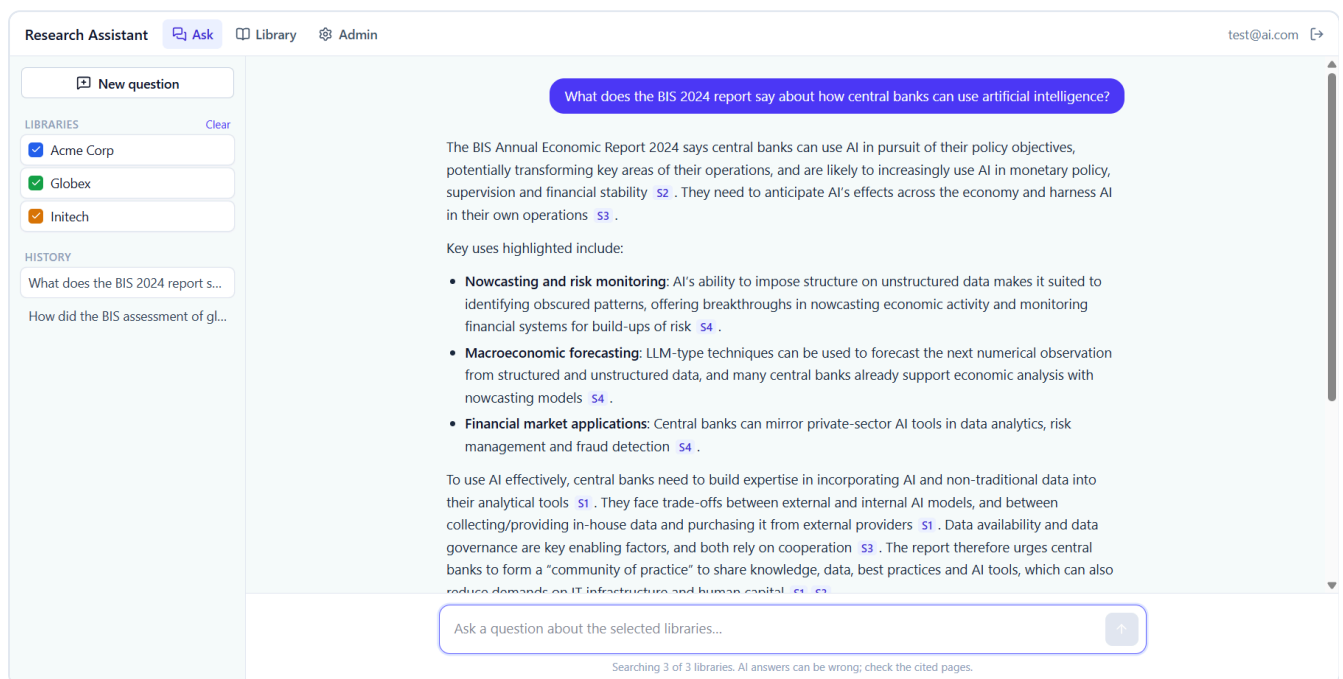


Figure 1. Asking a question in the prototype. The libraries to search are on the left. The answer cites four pages of the BIS Annual Economic Report 2024 ([S1]–[S4]). The question history is kept per user.

4 How the proposal meets the brief

Each requirement in the brief maps to a feature that runs in the prototype today.

REQUIREMENT	HOW IT IS ADDRESSED	STATUS
Ask questions across multiple reports	Library selector plus a chat interface. The Query Agent splits comparative or multi-period questions into up to three sub-queries, and retrieval caps pages per report so an answer draws on several reports. Follow-up questions keep the conversation's context.	Built
Ingest, index and retrieve efficiently	Direct-to-storage upload. Batched page-by-page extraction with pypdf. Header and footer cleanup. Overlapping chunks with embeddings. Hybrid vector and keyword search in Postgres, fused with Reciprocal Rank Fusion.	Built
AI answers grounded in the sources	The Answer Agent receives only the retrieved pages and must cite every factual sentence. The Evaluator Agent checks each claim; unsupported answers are rewritten once, then flagged if they still fail.	Built
Citations, document names and page references	Clickable [S#] chips show the report title and page. A source list sits under each answer. The PDF viewer opens at the cited <i>physical</i> page with the evidence highlighted, and shows the printed page number when it differs.	Built
Restrict access to authorised users and selected libraries	Invite-only Supabase Auth with two roles. Per-user library grants. Row-level security on every table and in the search function. Private PDF storage served through short-lived signed URLs.	Built
Workflow, architecture and stack	React front end; FastAPI back end; Supabase for Postgres with pgvector, Auth, Storage and Edge Functions; DeepSeek models; deployable to Vercel. See section 5.	Built
Evaluate quality, relevance and source accuracy	Live: an evaluator verdict and grounded score on every answer, plus thumbs-up/down feedback, shown on the Insights dashboard. Offline: a golden-set runner measuring Recall@10, MRR, citation accuracy, faithfulness, correctness and refusal accuracy.	Built Golden set next
Demonstrate customer value before an enterprise build	A 10-week client engagement with weekly meetings, measurable success criteria and a go/no-go decision. See section 13.	Proposed

5 Architecture and technology stack

The system has three tiers: a single-page web app, a stateless Python API, and Supabase as the managed data platform. The language models are called over an API, so no model infrastructure has to be hosted.

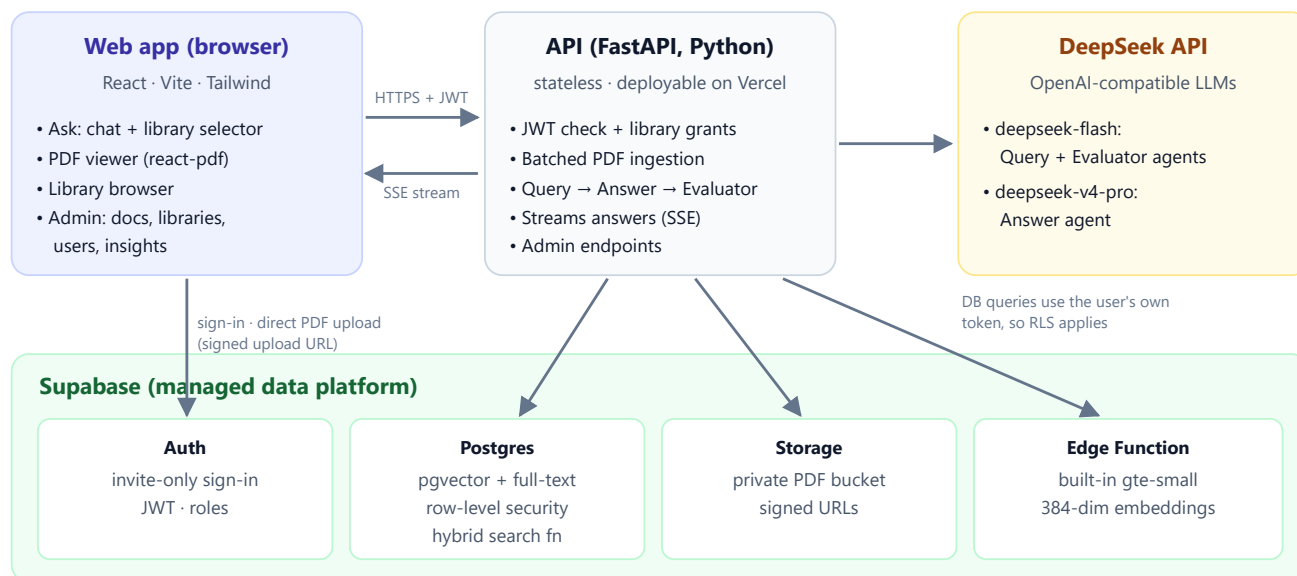


Figure 2. System architecture. The browser uploads PDFs straight to Storage, so large files never pass through the API. The API reads the database with the signed-in user's token, so the same security rules apply to every query.

Technology stack

LAYER	CHOICE	WHY
Front end	React 19, Vite, Tailwind CSS, TanStack Query, react-pdf	Fast to build and responsive. react-pdf exposes the text layer, which is used to highlight evidence.
API	FastAPI (Python), served on Vercel	Python has the best PDF and AI libraries, and FastAPI streams answers as server-sent events.
Language models	DeepSeek <code>deepseek-v4-pro</code> (answers), <code>deepseek-flash</code> (planning, evaluation)	OpenAI-compatible and low cost. The fast model keeps the per-answer check cheap. The model and reasoning effort are configurable per agent.
Embeddings	Supabase built-in <code>gte-small</code> via an Edge Function	No extra vendor or key. Can be swapped for a larger model in one file.
Database and search	Supabase Postgres, pgvector (HNSW), full-text search	Vectors, keywords, metadata and security rules in one database, so hybrid search runs in a single SQL function.
Auth and files	Supabase Auth and Storage	Invite-only accounts, JWTs and a private bucket, all managed.
Quality	pytest, Vitest, SQL security tests, golden-set evaluation, GitHub Actions CI	Every change is checked automatically; answer quality is measured, not assumed.

6 Ingestion, indexing and retrieval



Figure 3. Ingestion pipeline.

Reliable page references

Published reports often number pages with roman numerals in the front matter, or restart numbering in each section. The prototype therefore **does not rely on the printed page number**. It records every page by its physical position in the file (`page_index`), and every citation uses that number. The printed label is kept only for display. Of the 10 test reports collected (5 Federal Reserve, 5 BIS), 6 have printed page numbers that differ from their physical positions; the viewer shows both, e.g. "page 51 (printed 31)".

Efficient, resumable ingestion

- **Direct upload.** The browser uploads each PDF straight to Supabase Storage with a signed URL, which avoids API request-size limits.
- **Batched processing.** The API processes 25 pages per request and reports progress. A failed document can **resume** where it stopped, and re-running a batch is safe.
- **Text cleanup.** Headers and footers repeated on more than half the pages are removed, and words hyphenated across line breaks are rejoined.
- **Quality flags.** Pages with no extractable text, such as scanned images or blank dividers, are flagged in the admin panel.

Hybrid retrieval

- Each page is split into overlapping ~1,800-character chunks, the size the embedding model reads well. Every chunk keeps its page number.
- A single SQL function, `match_chunks`, runs two searches inside the database: **semantic similarity** (pgvector, HNSW index) and **keyword search** (Postgres full-text). It merges their rankings with **Reciprocal Rank Fusion**. Semantic search finds paraphrases; keyword search catches names, tickers and exact phrases.
- Hits are rolled up to pages. A page scores its best chunk for each sub-query, so evidence found by several sub-queries ranks higher. The top 8 pages go to the Answer Agent, with at most 4 per report so several reports can contribute.
- The Answer Agent gets the **whole page**, not a fragment, so it has full context.

7 Grounded answers: the multi-agent pipeline

Three specialised agents handle each question. Splitting the work keeps each prompt simple and makes the answer checkable.

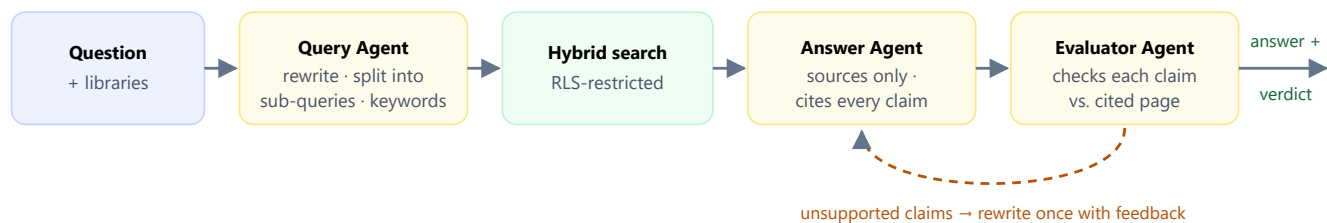


Figure 4. Question-answering pipeline. The answer streams to the user as it is written. The evaluator's verdict and evidence quotes are attached when it finishes.

AGENT	MODEL	RESPONSIBILITY
Query Agent	deepseek-flash (no reasoning, for speed)	Turns the question and chat history into a standalone question, 1–3 focused sub-queries for comparative or multi-period questions, and keywords for the keyword search.
Answer Agent	deepseek-v4-pro (low reasoning effort)	Answers only from the retrieved pages and cites every factual sentence. Points out changes between reports and periods. If the sources don't cover the question, it says so instead of guessing.
Evaluator Agent	deepseek-flash (low reasoning effort)	For every claim, records supported, partial or unsupported, whether the citation is correct, and a verbatim evidence quote. Produces a grounded score from 0 to 1. Citations to sources that don't exist are always marked wrong.

Quality gate. If any claim is unsupported, or the grounded score is below 0.75, the answer is rewritten **once** using the evaluator's findings and checked again. If it still fails, it is shown with a "Low confidence: verify before use" badge rather than hidden.

Research Assistant Ask Library Admin test@ai.com

New question

LIBRARIES Clear

- Acme Corp
- Globex
- Initech

HISTORY

What does the BIS 2024 report s...

How did the BIS assessment of gl...

Financial market applications: Central banks can mirror private-sector AI tools in data analytics, risk management and fraud detection ^{S4}.

To use AI effectively, central banks need to build expertise in incorporating AI and non-traditional data into their analytical tools ^{S1}. They face trade-offs between external and internal AI models, and between collecting/providing in-house data and purchasing it from external providers ^{S1}. Data availability and data governance are key enabling factors, and both rely on cooperation ^{S3}. The report therefore urges central banks to form a "community of practice" to share knowledge, data, best practices and AI tools, which can also reduce demands on IT infrastructure and human capital ^{S1} ^{S3}.

The report also flags challenges, such as new cyber risks that exploit weaknesses in LLMs to cause unintended behaviour or reveal sensitive information ^{S4}.

SOURCES

- ^{S2} BIS Annual Economic Report 2024 - page 112
- ^{S3} BIS Annual Economic Report 2024 - page 111
- ^{S4} BIS Annual Economic Report 2024 - page 18
- ^{S1} BIS Annual Economic Report 2024 - page 113

Grounded in sources · 100% ^

Every factual claim in the answer is directly supported by the cited BIS sources with correct citations. Every claim was supported by the cited page.

Ask a question about the selected libraries...

Searching 3 of 3 libraries. AI answers can be wrong; check the cited pages.

Figure 5. The evaluator's verdict is visible to the user. The source list sits under the answer, followed by the groundedness badge, which expands into the claim-by-claim check.

8 Citations and verification

Trust depends on users being able to check an answer quickly. The prototype makes verification a single click.

- Each `[S#]` marker in an answer is a clickable chip. Hovering shows the report title and page.
- A **Sources** list under every answer gives the report name and physical page for each citation, plus the printed page number when it differs.
- Clicking a citation opens the PDF in a side panel **at the cited page**. The evaluator's evidence quote is **highlighted** on the page and scrolled into view.
- Users can page through the document, zoom, return to the cited page, or open the original PDF in a new tab.

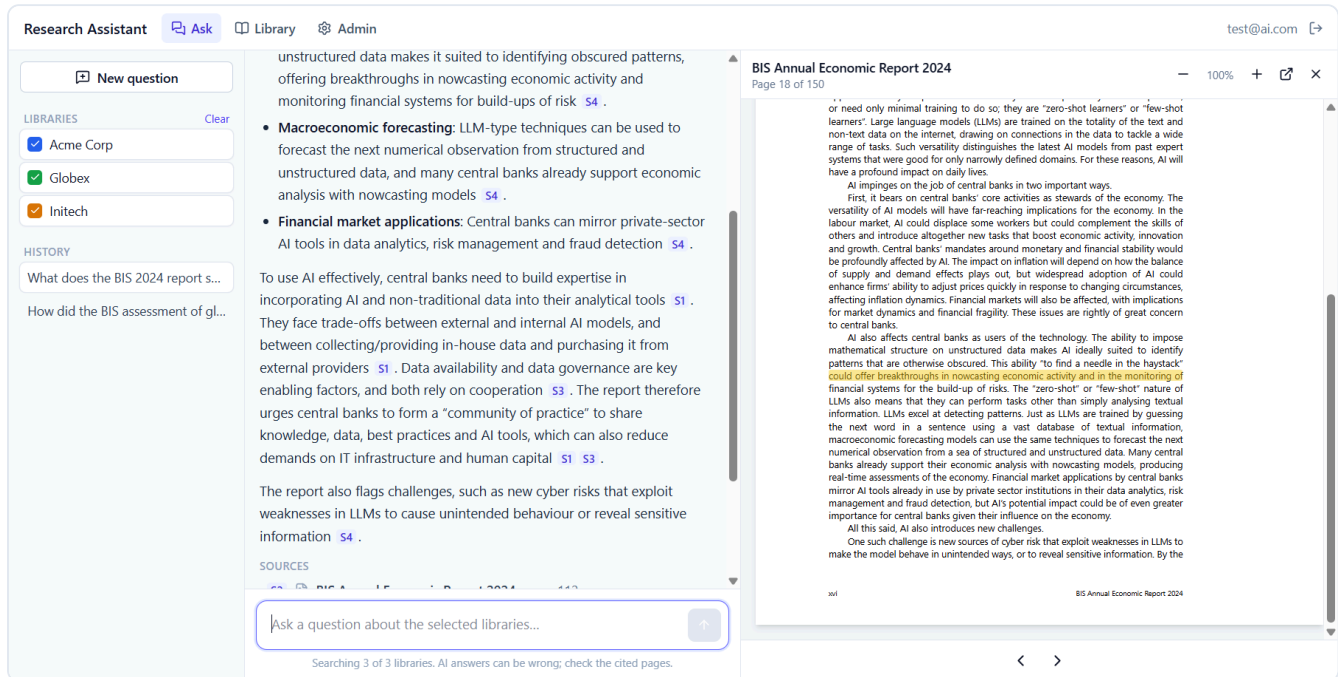


Figure 6. One-click verification. Clicking citation S4 opens the BIS Annual Economic Report 2024 at page 18 of 150 and highlights the sentence the claim relies on: "...could offer breakthroughs in nowcasting economic activity and in the monitoring of...".

9 Access control

Restricting each user to the research libraries they are entitled to is enforced at several layers, so a bug in one layer doesn't expose data.

LAYER	CONTROL
Identity	Supabase Auth with email and password or a one-time link. Accounts are invitation-only ; there is no public sign-up.
Roles and grants	Users are <code>admin</code> or <code>user</code> . Admins grant each user specific libraries, and the label on a report decides which libraries it belongs to.
Database (main control)	Postgres row-level security on documents, pages, chunks, libraries, conversations and messages. A user sees a document only if it carries a library they hold. The search function runs with the caller's permissions, so naming another library's id still returns nothing.
API	Verifies the user's token, intersects the requested libraries with the user's grants before any search, and queries the database with the user's own token.
Files	PDFs sit in a private bucket. A short-lived signed URL is issued only after the database confirms the user can see the document.
Privacy	Conversations are private to their owner. Administrators can read answers in Insights, so they can review quality.

Tested. 24 automated database checks run the migrations against a real Postgres with pgvector. They confirm that users cannot see other libraries' documents, chunks or search results, cannot promote themselves or grant themselves access, and cannot read other users' conversations.

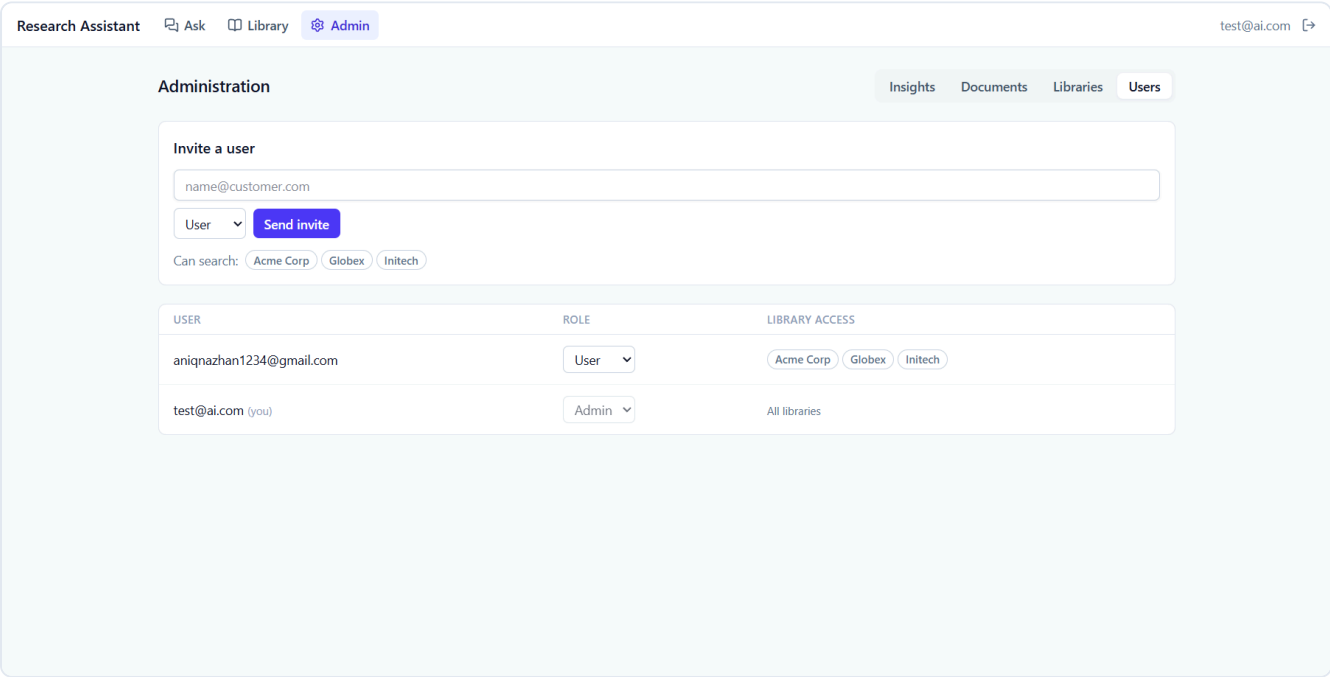


Figure 7. User management. Admins invite users, set their role, and choose which libraries each one can search.

10 Administration

Administrators run the libraries without technical help.

- **Documents:** drag-and-drop upload with per-file progress, library tagging, status (processing, ready or failed) with error details, resume, re-index and delete, and warnings for pages without text.
- **Libraries:** create, rename and colour libraries; see document and user counts; bulk-assign reports.
- **Users:** invite, set roles, and grant libraries, as shown in figure 7.
- **Insights:** documents indexed, questions asked, average groundedness, answers revised by the evaluator, the helpful rate, and a filterable question log with latency and token usage.

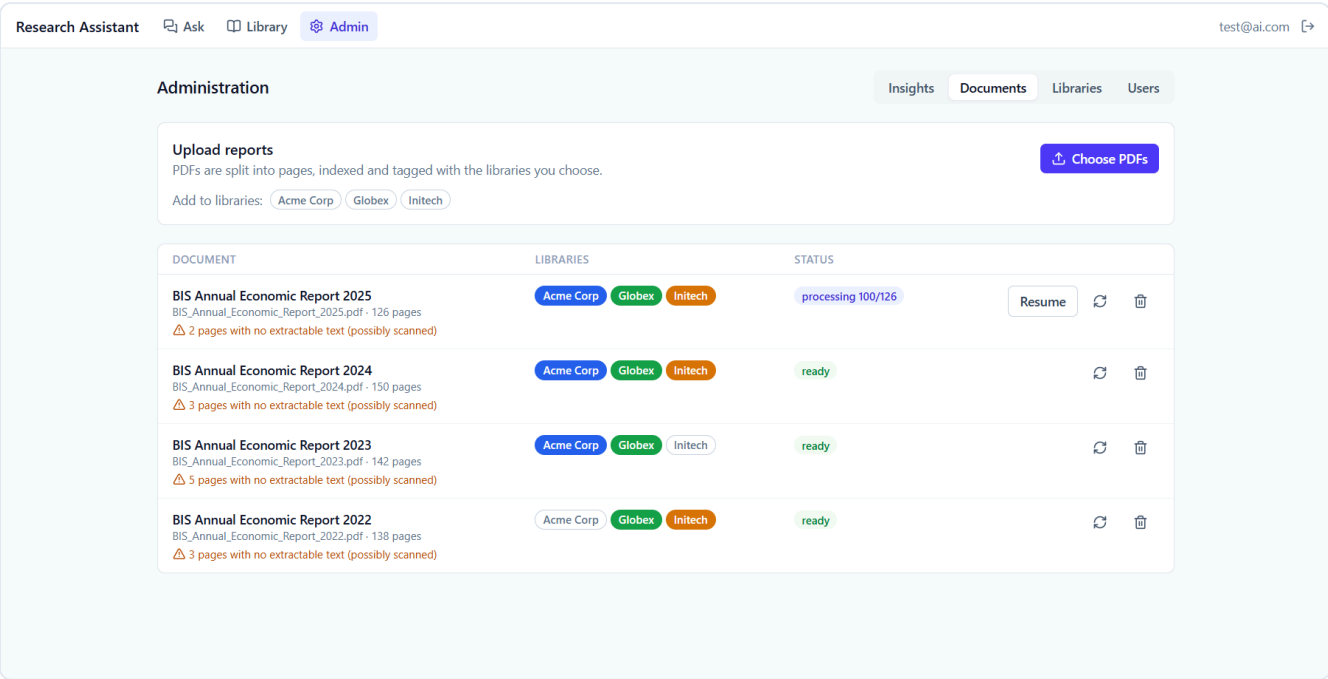


Figure 8. Document management. Real BIS Annual Economic Reports (2022–2025), tagged with libraries. The 2025 report shows resumable batch indexing in progress (100 of 126 pages).

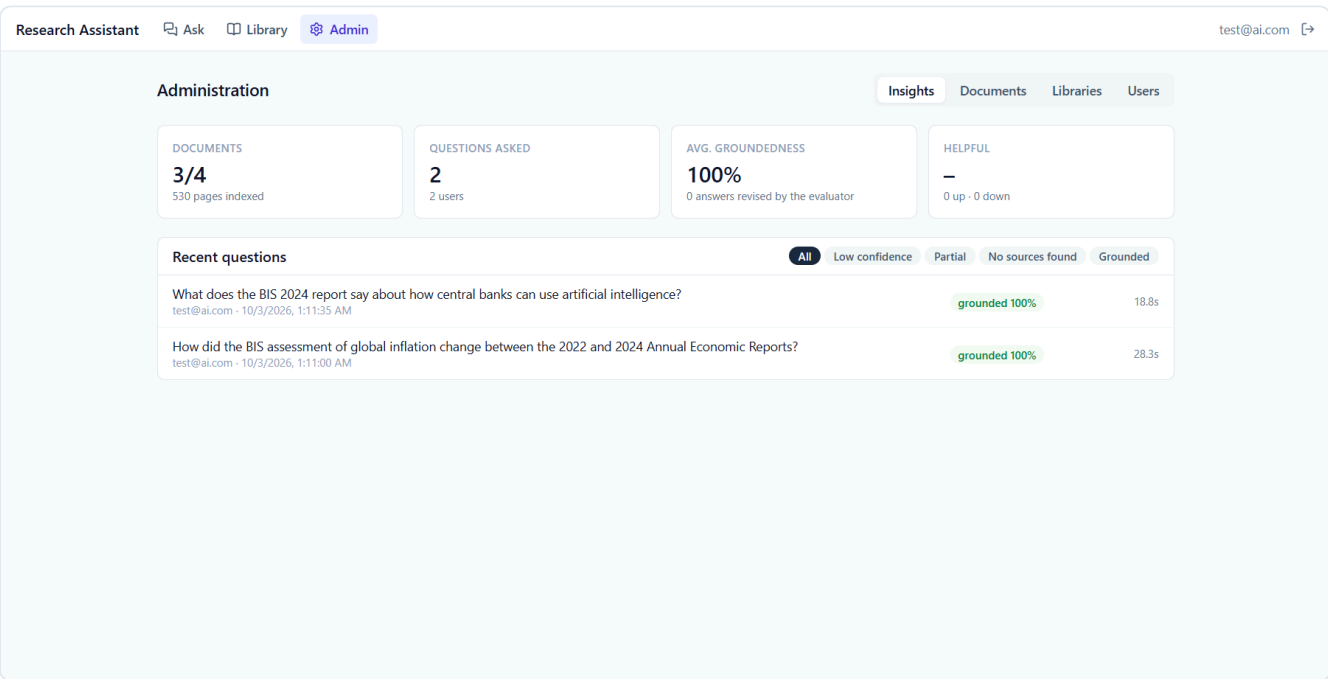


Figure 9. Insights dashboard. The evidence base for a go/no-go decision: groundedness, feedback, and every question with its verdict and latency.

11 Evaluating answer quality

Quality is measured in two complementary ways: continuously on every live answer, and offline against an expert-written test set before any change ships.

Live, on every answer

- The Evaluator Agent's verdict, grounded score and claim checks are stored with the answer.
- Users rate answers with thumbs up or down and an optional note.
- Insights tracks the trends and lists low-confidence and no-source questions. These show gaps in the library or in retrieval.

Offline, against a golden set

A domain expert writes 30–50 questions across four types: single fact, cross-report synthesis, comparison over time, and unanswerable (the assistant must decline). Each question records the expected report pages and a reference answer. The included runner (`evals/run_eval.py`) sends them through the same pipeline the app uses and reports:

METRIC	WHAT IT MEASURES	TARGET
Recall@10, MRR	Relevance of retrieval: were the expected pages found, and how high did the first one rank?	≥ 0.85
Citation accuracy	Source accuracy: does each cited page actually contain the claim? Scored by an LLM judge, with about 20 verdicts checked by hand.	≥ 0.90
Faithfulness	Share of claims supported by the cited sources.	track
Correctness, relevance	Answer quality against the expert's reference answer, on a 1–5 scale.	track
Refusal accuracy	Unanswerable questions are declined, not answered.	≥ 0.90
Latency, tokens	Time to answer (p50/p95) and cost per question.	track

Reports are saved as CSV and markdown, so runs before and after a change to prompts, chunking or retrieval can be compared side by side.

12 Prototype status and evidence

4

BIS reports uploaded (3 fully indexed)

530

pages indexed · 1,215 chunks

≈ 9k / 4k

input / output tokens per question

< US\$0.04

model cost per question at DeepSeek peak list prices

What works today

CAPABILITY	STATUS
Invite-only sign-in, admin and user roles, per-user library grants	Working
PDF upload, batched indexing with progress, resume and re-index	Working
Hybrid semantic and keyword search, restricted by database security	Working
Three-agent answering with streaming, citations and an evaluator quality gate	Working
PDF viewer that opens at the cited page and highlights the evidence	Working
Admin panel: documents, libraries, users, insights	Working
Automated tests: 38 API, 7 front end, 24 database security; CI pipeline	Passing
Golden-set evaluation runner	Built
Golden set written with a domain expert and baseline scores	Next step

Findings from early testing

Strong single-report answer. "What does the BIS 2024 report say about how central banks can use artificial intelligence?" produced a structured answer citing four pages (pages 18, 111, 112 and 113). The evaluator rated it 100% grounded, and the evidence was highlighted correctly in the viewer (figures 1, 5 and 6). Time: 18.8 s.

Retrieval weakness found. For a broad comparative question about inflation across the 2022 and 2024 reports, search returned mainly **reference lists and endnotes**. Those pages are dense with titles containing the search terms. The assistant correctly said the sources didn't contain the assessment instead of inventing one, but the right pages were missed. **Fix planned:** detect bibliography and endnote pages at ingestion and exclude or down-weight them in search, then confirm with the golden set.

These are early, qualitative observations from a small number of questions. They are not yet a measured benchmark; the golden-set runs during the client engagement will provide one.

13 Proving customer value: 10-week client engagement

The prototype is ready to put in front of the client. Over **10 weeks** we hold a **weekly meeting with the client**: we show progress in the live prototype, review results together, and agree the priorities for the next week. Every week is shaped by the client's feedback, and the engagement answers the question that matters before a larger investment: **do customers get faster, trustworthy answers from the research they already pay for?**

WEEK	WEEKLY CLIENT MEETING	WORK BETWEEN MEETINGS
1	Kick-off: agree goals, success criteria, the research libraries in scope and the users taking part.	Receive the client's reports and set up the client's environment.
2	Walk through the prototype on the client's own reports. Review the first answers together.	Load 1–2 libraries. Fix the reference-page retrieval issue.
3	Agree the golden set of test questions with the client's domain expert.	Write the golden set and record baseline quality scores.
4	Review baseline scores. Agree the access rules: which users see which libraries.	Tune retrieval and prompts. Configure user accounts and library access.
5	Onboard 5–10 client users with a short walkthrough.	Run a timed task study: real questions answered manually versus with the assistant.
6	Review the timed-task results and early user feedback.	Act on feedback. Re-run the golden set after each change.
7	Review Insights: low-confidence answers, questions with no sources, ratings.	Close gaps in the library and in retrieval. Add reports the client requests.
8	Demo the improvements. Agree any final adjustments.	Final tuning. Golden-set run against the targets.
9	Short interviews with the client's users. Review results against the success criteria.	Compile results, costs and the production options.
10	Final review: results, recommendation and the go/no-go decision on a production build.	Hand over the final report and the prototype.

Success criteria for "go" (reviewed in weeks 9–10)

MEASURE	TARGET
Time to answer a research question versus manual PDF search	≥ 50% faster
Citation accuracy (golden set)	≥ 90%
Retrieval Recall@10 (golden set)	≥ 85%
Unanswerable questions correctly declined	≥ 90%
Answers rated helpful by client users	≥ 70%
Client users who would use it weekly	≥ 60%

Demo script (3–5 minutes)

1. Pick a library and ask a single-report question. Click a citation and show the highlighted evidence.
2. Ask a question that spans several reports and show citations from each.
3. Ask something the library doesn't cover and show the assistant declining.
4. Admin: restrict a user to one library, then show the other library disappear for them.
5. Insights: show groundedness and feedback as the basis for the decision.

14 Risks, limits and the path to production

Known limits of the prototype

LIMIT	IMPACT AND MITIGATION
Reference and endnote pages can crowd out body text	Seen in testing. Exclude or down-weight them at ingestion (engagement week 2).
Embedding model is small and English-only (gte-small)	Hybrid keyword search compensates. Swapping to a larger or multilingual model changes one file.
Scanned PDFs, tables and charts	Pages without a text layer are flagged but not indexed. Tables are read as plain text. Add OCR and layout-aware extraction in production.
Response time of 19–28 s	The answer streams as it is written, so text appears sooner. The evaluator check and model reasoning add time; reasoning effort is configurable per agent.
Evaluator judges itself generously at times	An honest "not covered" reply was rated 100% grounded. Spot-check judge verdicts by hand and consider a judge from a different model family for offline evaluation.
Data sent to a third-party model API	Only retrieved excerpts are sent, never whole libraries. Confirm the provider's data terms, or use an approved provider or region, before using sensitive research.

What a production build would add

- OCR and layout-aware extraction for tables and figures
- A background ingestion queue for large volumes
- A re-ranking step after hybrid search
- SSO/SAML and automated user provisioning (SCIM)
- Audit logging of document access and answers
- Per-tenant isolation for each customer
- Usage metering, quotas and rate limits
- Data-retention and deletion controls

Conclusion. The concept is no longer theoretical. A working prototype already answers questions across real research reports with verifiable, page-level citations and database-enforced access control. A 10-week engagement with weekly client meetings is a low-cost way to measure its value before committing to a full enterprise implementation.